Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
Faculty: Prof. Karthika V	**Slot:** L43 + L44
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Week 1 – Caesar Cipher

Code:

Sender:

```
import java.util.*;

import java.io.*;

import java.net.*;

public class Caesar_Sender {

    private Socket socket = null;

    private DataInputStream input = null;

    private DataOutputStream output = null;

    private Scanner sc = null;

    public Caesar_Sender(String address, int port) {

        try {

            socket = new Socket(address, port);

            System.out.println("Client is connected!");

            input = new DataInputStream(socket.getInputStream());

            output = new DataOutputStream(socket.getOutputStream());

            sc = new Scanner(System.in);

        } catch (UnknownHostException u) {

            System.out.println(u);

        } catch (IOException i) {

            System.out.println(i);

        }

    }

    String caesarEncrypt(String pt, int k) {
```

```
    StringBuilder ct = new StringBuilder();
    for (char c : pt.toCharArray()) {
        if (Character.isUpperCase(c)) {
            int ch = (int) c - 'A';
            ch = (ch + k) % 26;
            char cs = (char) (ch + 65);
            ct.append(cs);
        } else if (Character.isLowerCase(c)) {
            int ch = (int) c - 'a';
            ch = (ch + k) % 26;
            char cs = (char) (ch + 97);
            ct.append(cs);
        } else {
            ct.append(c);
        }
    }
    return ct.toString();
}
public void Work() {
    try {
        System.out.print("Enter the message: ");
        String[] ips = sc.nextLine().trim().split("\\s+");
        System.out.print("Enter the key: ");
        int k = sc.nextInt();
        Vector<String> ct = new Vector<>();
        for (String word : ips) {
            ct.add(caesarEncrypt(word, k));
        }
        String cts = String.join(" ", ct);
        System.out.println("The encrypted text is: " + cts);
        output.writeInt(k);
```

```
            output.writeUTF(cts);
            System.out.println(input.readUTF());
            input.close();
            output.close();
            sc.close();
        } catch (IOException i) {
            System.out.println(i);
        }
    }
    public static void main(String[] args) {
        Caesar_Sender cc = new Caesar_Sender("127.0.0.1", 5431);
        cc.Work();
    }
}
```

Receiver:

```
import java.util.*;
import java.io.*;
import java.net.*;
public class Caesar_Receiver {
    private ServerSocket server = null;
    private Socket socket = null;
    private DataInputStream input = null;
    private DataOutputStream output = null;
    public Caesar_Receiver(int port) {
        try {
            server = new ServerSocket(port);
            socket = server.accept();
            System.out.println("Client is connected!\n");
            input = new DataInputStream(socket.getInputStream());
            output = new DataOutputStream(socket.getOutputStream());
        } catch (UnknownHostException u) {
```

```
        System.out.println(u);
    } catch (IOException i) {
        System.out.println(i);
    }
}
String caesarDecrypt(String ct, int k) {
    StringBuilder pt = new StringBuilder();
    for (char c : ct.toCharArray()) {
        if (Character.isUpperCase(c)) {
            int ch = (int) c - 'A';
            ch = (ch - k) % 26;
            if (ch < 0) {
                ch += 26;
            }
            char cs = (char) (ch + 65);
            pt.append(cs);
        } else if (Character.isLowerCase(c)) {
            int ch = (int) c - 'a';
            ch = (ch - k) % 26;
            if (ch < 0) {
                ch += 26;
            }
            char cs = (char) (ch + 97);
            pt.append(cs);
        } else {
            pt.append(c);
        }
    }
    return pt.toString();
}
public void Work() {
```

```
        try {
            int k = input.readInt();
            String ip = input.readUTF();
            String[] ips = ip.trim().split("\\s+");
            Vector<String> ct = new Vector<>();
            for (String word : ips) {
                ct.add(caesarDecrypt(word, k));
            }
            System.out.println("The received encrypted text is: " + ip);
            System.out.println("The received key is: " + k);
            String cts = String.join(" ", ct);
            System.out.println("The decrypted text is: " + cts);
            output.writeUTF("Success from receiver!");
            input.close();
            output.close();
        } catch (IOException i) {
            System.out.println(i);
        }
    }
    public static void main(String[] args) {
        Caesar_Receiver cs = new Caesar_Receiver(5431);
        cs.Work();
    }
}
```

Output:

Sender:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1> java .\Caesar_Sender.java
Client is connected!
Enter the message: Meet me.
Enter the key: 3
The encrypted text is: Phhw ph.
Success from receiver!
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1>
```

Receiver:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1> java .\Caesar_Receiver.java
Client is connected!

The received encrypted text is: Phhw ph.
The received key is: 3
The decrypted text is: Meet me.
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1>
```

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